MATH 345 Skeleton Notes
Unit 6: Constraints and SVD
Section 6.4: Singular value decomposition

The previous section found singular values from $A^T A$. This section packages the same information into one factorization. The right singular vectors are special input directions. The singular values are stretch factors. The left singular vectors are the corresponding output directions. The singular value decomposition is the rectangular analogue of orthogonal diagonalization. Orthogonal diagonalization applies to symmetric square matrices. SVD applies to every real matrix.

Definition: Singular value decomposition Let A be an $m \times n$ matrix, and let $s = \min(m, n)$. A singular value decomposition, or SVD, of A is a factorization

$$A = U\Sigma V^T,$$

where U is an $m \times m$ orthogonal matrix, V is an $n \times n$ orthogonal matrix, and Σ is an $m \times n$ diagonal matrix whose diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_s \geq 0$$

are nonnegative.
The numbers σ_i are the singular values of A. The columns

$$\mathbf{u}_1, \dots, \mathbf{u}_m$$

of U are left singular vectors. The columns

$$\mathbf{v}_1, \dots, \mathbf{v}_n$$

of V are right singular vectors.
If r is the number of positive singular values, then

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i \qquad \text{for } 1 \leq i \leq r,$$

and

$$A\mathbf{v}_i = \mathbf{0} \qquad \text{for } i > r.$$

Note: Notation for SVD Some references write the same factorization as

$$A = P\Sigma Q^T.$$

In this book we use

$$A = U\Sigma V^T$$

because this matches the NumPy output names `U, s, Vt = np.linalg.svd(`
The roles are the same: U stores left singular vectors, V stores right singular vectors, and Σ stores singular values.

The factorization also gives a coordinate formula. If $r = \operatorname{rank}(A)$, then

$$A\mathbf{x} = \sum_{i=1}^{r} \sigma_i (\mathbf{v}_i \cdot \mathbf{x}) \mathbf{u}_i.$$

The dot product $\mathbf{v}_i \cdot \mathbf{x}$ measures how much of the input points in the i-th right singular direction. The singular value σ_i stretches that coordinate. The vector $\mathbf{u}_i$ gives the output direction.
An $m \times n$ matrix A maps the unit sphere (the set of all unit vectors) in $\mathbb{R}^n$ to an ellipsoid in $\mathbb{R}^m$ (i.e., a sphere which has been stretched and compressed along different

principal axes). The singular value decomposition identifies the principal axes of this ellipsoid and the corresponding scaling factors. textbook figure fig-unit-sphere-to-ellipse gives an example of how this may be visualized when A is a 2×3 matrix.

Lemma: Properties of $A^T A$ and AA^T Let A be an $m \times n$ matrix. Then:

1. The eigenvalues of $A^T A$ and AA^T are real and non-negative.
2. $A^T A$ and AA^T have the same set of positive eigenvalues.

Proof 1. The matrices $A^T A$ and AA^T are symmetric, so by Principal Axes Theorem their eigenvalues are real. If $A^T A\mathbf{v} = \lambda\mathbf{v}$ for $\mathbf{v} \neq \mathbf{0}$, we calculate that

$$\|A\mathbf{v}\|^2 = (A\mathbf{v})^T(A\mathbf{v}) = \mathbf{v}^T A^T A\mathbf{v} = \mathbf{v}^T\lambda\mathbf{v} = \lambda\|\mathbf{v}\|^2$$

Since $\|A\mathbf{v}\|^2 \geq 0$ and $\|\mathbf{v}\|^2 > 0$, we conclude that $\lambda \geq 0$.
2. Let λ be a positive eigenvalue of $A^T A$ with eigenvector $\mathbf{v} \neq \mathbf{0}$. Then $A^T A\mathbf{v} = \lambda\mathbf{v}$. Let $\mathbf{w} = A\mathbf{v}$. Note that $\mathbf{w} \neq \mathbf{0}$ since $\lambda > 0$. We have:

$$AA^T\mathbf{w} = AA^T(A\mathbf{v}) = A(A^T A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda\mathbf{w}.$$

So λ is also an eigenvalue of AA^T. The argument works in reverse by replacing A with A^T, proving that $A^T A$ and AA^T have the same positive eigenvalues.

Theorem: Computing the Singular Value Decomposition Given an $m \times n$ matrix A, one way to obtain an SVD

$$A = U\Sigma V^T$$

is as follows.

1. Find an orthonormal eigenbasis

$$\mathbf{v}_1, \ldots, \mathbf{v}_n$$

for $A^T A$. Let λ_i be the eigenvalue associated with $\mathbf{v}_i$. Order the vectors so that

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$$

and $\lambda_i = 0$ for $i > r$.

2. Define

$$\sigma_i = \sqrt{\lambda_i}.$$

3. For $1 \leq i \leq r$, define

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i.$$

4. Extend

$$\mathbf{u}_1, \ldots, \mathbf{u}_r$$

to an orthonormal basis

$$\mathbf{u}_1, \ldots, \mathbf{u}_m$$

of $\mathbb{R}^m$.

5. Set

$$V = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n], \qquad U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_m],$$

and let Σ be the $m \times n$ diagonal matrix with diagonal entries $\sigma_1, \ldots, \sigma_r$ followed by zeros.

Then

$$A = U\Sigma V^T.$$

Proof For $i \neq j$ with $1 \leq i, j \leq r$,

$$\begin{aligned}
&\mathbf{u}_i \cdot \mathbf{u}_j \\
&\quad = \frac{1}{\sigma_i \sigma_j}(A\mathbf{v}_i) \cdot (A\mathbf{v}_j) \\
&\quad = \frac{1}{\sigma_i \sigma_j}\mathbf{v}_i^T A^T A\mathbf{v}_j..
\end{aligned}$$

Since $A^T A\mathbf{v}_j = \lambda_j \mathbf{v}_j$, this becomes

$$\frac{\lambda_j}{\sigma_i \sigma_j}(\mathbf{v}_i \cdot \mathbf{v}_j) = 0.$$

Also,

$$\begin{aligned}
&\|\mathbf{u}_i\|^2 \\
&\quad = \frac{1}{\sigma_i^2}\|A\mathbf{v}_i\|^2 \\
&\quad = \frac{1}{\lambda_i}\mathbf{v}_i^T A^T A\mathbf{v}_i \\
&\quad = 1.
\end{aligned}$$

Thus $\mathbf{u}_1, \ldots, \mathbf{u}_r$ are orthonormal.
If $i > r$, then $\lambda_i = 0$, so

$$\begin{aligned}
&\|A\mathbf{v}_i\|^2 \\
&\quad = \mathbf{v}_i^T A^T A\mathbf{v}_i \\
&\quad = 0.
\end{aligned}$$

Therefore $A\mathbf{v}_i = \mathbf{0}$.
Now compare the columns of AV and $U\Sigma$:

$$\begin{aligned} AV & \\ &= [A\mathbf{v}_1 \ \cdots \ A\mathbf{v}_n] \\ &= [\sigma_1\mathbf{u}_1 \ \cdots \ \sigma_r\mathbf{u}_r \ \mathbf{0} \ \cdots \ \mathbf{0}] \\ &= U\Sigma. \end{aligned}$$

Since V is orthogonal, $V^{-1} = V^T$. Multiplying $AV = U\Sigma$ on the right by V^T gives

$$A = U\Sigma V^T.$$

The same computation also shows that the image of A is spanned by

$$\mathbf{u}_1, \dots, \mathbf{u}_r,$$

so $r = \operatorname{rank}(A)$.

Note: Non-uniqueness of the SVD The SVD is not unique. A matched pair of singular vectors can be multiplied by -1 without changing the product:

$$\sigma_i\mathbf{u}_i\mathbf{v}_i^T = \sigma_i(-\mathbf{u}_i)(-\mathbf{v}_i)^T.$$

When a singular value is zero, there can also be many valid choices for the corresponding orthonormal basis vectors. The roles are more important than the specific signs.

Here is one complete SVD computation, broken into the same steps as the theorem.

Activity: Computing an SVD, part 1: right singular vectors

Let

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}.$$

1. Compute $A^T A$.
2. Find the eigenvalues of $A^T A$.
3. Find an orthonormal eigenbasis

   $$\mathbf{v}_1, \mathbf{v}_2$$

   for $A^T A$, with the positive eigenvalue first.
4. Find the singular values of A.

Activity: Computing an SVD, part 2: output directions

Use the vectors from the previous activity.

1. Compute $A\mathbf{v}_1$.
2. Compute $A\mathbf{v}_2$.
3. Compute

   $$\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1.$$

4. What does $A\mathbf{v}_2 = \mathbf{0}$ say about the matrix map?

Activity: Computing an SVD, part 3: assembling the factorization

The SVD computation needs an orthonormal basis of $\mathbb{R}^3$ beginning with

$$\mathbf{u}_1 = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}.$$

Use the following completion:

$$\mathbf{u}_2 = \begin{bmatrix} (2/3)\sqrt{2} \\ (1/6)\sqrt{2} \\ -(1/6)\sqrt{2} \end{bmatrix}, \qquad \mathbf{u}_3 = \begin{bmatrix} 0 \\ (1/2)\sqrt{2} \\ (1/2)\sqrt{2} \end{bmatrix}.$$

1. Verify that $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are orthonormal.
2. Form U, Σ, and V.
3. Explain why $AV = U\Sigma$.
4. Conclude that $A = U\Sigma V^T$.

It turns out that any singular value decomposition contains a great deal of information about an $m \times n$ matrix A and the subspaces associated with A.

Definition: Fundamental Subspaces The four fundamental subspaces corresponding to a $m \times n$ matrix A are:

$$\mathrm{row}(A), \qquad \mathrm{col}(A), \qquad \mathrm{null}(A), \qquad \mathrm{null}(A^T).$$

More explicitly,

$$\mathrm{row}(A) = \mathrm{span}\{\text{rows of } A\},$$

$$\mathrm{col}(A) = \mathrm{span}\{\text{columns of } A\},$$

$$\mathrm{null}(A) = \{\mathbf{x} \in \mathbb{R}^n \, : \, A\mathbf{x} = \mathbf{0}\},$$

and

$$\mathrm{null}(A^T) = \{\mathbf{y} \in \mathbb{R}^m \, : \, A^T\mathbf{y} = \mathbf{0}\}.$$

Theorem: Relations Between Singular Value Decomposition and Fundamental Subspaces Let A be an $m \times n$ matrix, and let

$$A = U\Sigma V^T$$

be an SVD. Suppose $r = \mathrm{rank}(A)$, and write

$$U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_m], \qquad V = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n].$$

Then:

- $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is an orthonormal basis for $\mathrm{col}(A)$.
- $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$ is an orthonormal basis for $\mathrm{null}(A^T)$.
- $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is an orthonormal basis for $\mathrm{row}(A)$.
- $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$ is an orthonormal basis for $\mathrm{null}(A)$.

Activity: Reading the computed SVD

Use the SVD computed above:

$$A = U\Sigma V^T,$$

where

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$

$$U = \begin{bmatrix} 1/3 & (2/3)\sqrt{2} & 0 \\ -2/3 & (1/6)\sqrt{2} & (1/2)\sqrt{2} \\ 2/3 & -(1/6)\sqrt{2} & (1/2)\sqrt{2} \end{bmatrix},$$

and

$$\Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

1. What is $\mathrm{rank}(A)$?
2. Give an orthonormal basis for $\mathrm{row}(A)$.
3. Give an orthonormal basis for $\mathrm{null}(A)$.
4. Give an orthonormal basis for $\mathrm{col}(A)$.
5. Give an orthonormal basis for $\mathrm{null}(A^T)$.
6. Which input direction is forgotten?
7. Which input direction is transmitted, and by what stretch factor?

Note: SVD as an information channel The singular value decomposition

$$A = U\Sigma V^T$$

can be read as an anatomy of the matrix map $\mathbf{x} \mapsto A\mathbf{x}$. Read from right to left:

- V^T expresses the input in special input directions.
- Σ stretches, shrinks, or erases those directions.
- U expresses the result in special output directions.

Large singular values correspond to directions that are strongly transmitted. Small singular values correspond to directions that are weakly transmitted. Zero singular values correspond to forgotten directions. This is a precise version of the Unit 2 question: what information does a matrix map keep, weaken, or forget?